VARSHAA SEKAR JAYANTHI

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EDUCATION

Northwestern University, McCormick School of Engineering — Chicago, IL
Master of Science in Information Technology – GPA: 3.75/4.0 — *Sep 2025 – Dec 2026*
- **Coursework:** Computer Networks, Machine Learning, Deep Learning, Finance, Software Architecture, Algorithms.
- **Honors:** Advanced to the prototype round, Longaeva AI Hackathon (hedge fund, NYC).

Carnegie Mellon University — Pittsburgh, PA
Bachelor of Science in Information Systems – Minor in Business Administration — *Aug 2019 – May 2023*
- **Coursework:** Application Development, Databases, Data Structures, Data Science, Statistics, Information Security and Assurance.
- **Honors:** Grace Hopper Celebration Scholarship; First prize: Statistical Graphics Challenge; Senior Leadership Recognition Award.

EXPERIENCE

Advantage Futures — Chicago, IL
AI Engineer Consultant, Northwestern Forge – AI knowledge hub for an FCM (futures brokerage) — *Sep 2026 – Present*
- Built a retrieval-augmented generation (RAG) knowledge hub using Amazon S3 for document storage and open-source models for source-cited internal document Q&A; delivered a successful demo beyond Phase 1.
- Consolidated the firm's unstructured policies and procedures into a unified retrieval pipeline, reducing information-sourcing time by 75%.
- Lead a 4-person team in weekly sessions with the firm's founder and CEO, translating workflows into architecture and scope.

Tenacitics, Inc. — Houston, TX
Software Engineer Intern – Internal SQL deployment platform, 4 environments – Python, PostgreSQL, React — *Jun 2026 – Aug 2026*
- Built an internal SQL deployment platform end to end, replacing a manual 3-system ticket-and-email handoff and cutting release effort 68% across 120 developers and DBAs.
- Engineered transactional deployment so a failed script leaves the database unchanged, removing partial-migration cleanup.
- Managed a suite of 113 tests at 98% coverage against a live PostgreSQL database, spanning end-to-end and regression paths.

Northwestern University — Chicago, IL
Graduate Researcher, Agentic Systems Evaluation – Python, Kubernetes, Docker — *Jan 2026 – Jun 2026*
- Evaluated and refined a multi-agent system which triages and fixes Kubernetes and Docker deployment failures.
- Ran ablation testing on GPT-5.5 and Claude Sonnet across scenarios, raising task accuracy 54% to 78.5% in 400+ evaluations.
- Engineered the VERA reinforcement loop to resolve environment precondition failures, the dominant failure mode over model reasoning.

Computer Enterprises Inc. — Pittsburgh, PA
Software Developer, vendor-side owner – Client: Zelis Payments, large-scale healthcare payments — *Jun 2024 – Aug 2025*
- Shipped 10 major features across claims pricing and payment workflows in C#/.NET and React, reducing the bug backlog 35%.
- Improved database performance 40% by rewriting slow queries, restructuring joins, adding indexes and narrowing full-table scans.
- Automated verification of 10+ enterprise data migrations with an LLM pipeline at 98% accuracy and 20% lower cycle time.

UPMC Corporate Services — Pittsburgh, PA
Software Engineer – C#/.NET, ASP.NET MVC, MS SQL Server, Azure DevOps — *Jul 2023 – Jun 2024*
- Developed roughly 20 features for a $1.5M Health Plan web app presented to the CTO and owned the lower-environment database.
- Instrumented a triage tracker across 10 environments, mapping failing tests to owners; presented improvements to C-Suite.
- Lowered production error rates 47% with Cypress and Playwright regression suites gating every release through CI/CD.

Carnegie Mellon University — Pittsburgh, PA
Teaching Assistant – Application Design and Development — *Jan 2023 – Aug 2023*
- Led weekly labs for 80+ students in a core CMU course: MVC, relational database design, and full-stack Rails.

PROJECTS

Matching Engine for Prediction Markets and Market-Making Simulator | *Python; C++17 port in progress* — May 2026 – Present
- Engineered the matching engine: price-time priority, partial fills, multi-level walking, benchmarked at 380,000 operations/sec.
- Designed the market maker and counterparty agents on a simulation engine: 5 switchable quoting mechanisms, Poisson arrivals.
- Reconciled spread capture, inventory drift and fees against realized cash; 11 tests cover matching and harness invariants.
- Monitored a live market data feed of 478 contracts; mid-price screening reports 38x more violations than are executable.

IMC Prosperity 4 Global Algorithmic Trading Competition | *Python* — Apr 2026 – May 2026
- Team captain, team of two; reached the final round, ranking 1,275 of 18,803 teams (top 7%).
- Doubled per-day profit with a rolling-mean fair value; a regression signal with slope-based sizing beat a moving-average baseline 35%.
- Priced the options round with Black-Scholes and a Newton-Raphson implied volatility solve on vega, quoting off the anchor's live IV.
- Screened 100 candidate pairs for statistical arbitrage with PCA across a 50-product universe, PC1 explaining 12.7% of variance.

BluePrint | *Software Developer, early-stage startup AI SDLC governance platform; TypeScript, PostgreSQL, React* — Jul 2026 – Sep 2026
- Rewrote a recursive CTE after diagnosing fan-in convergence, cutting a graph query 31s to 183ms and 472,421 rows to 123.
- Closed 2 live cross-tenant data-exposure paths, one a state-changing write, by root-causing parameter loss in Express middleware.

LEADERSHIP

Carnegie Mellon Student Government, both positions elected, 2021 to 2023:
- Lead Financial Advisor, Joint Funding Committee: reviewed organization financial statements and set line-item allocations from a $2M annual pool against $4.2M requested by 250 organizations, defending cuts at appeals; led 3 advisors.
- Vice President, Activities Board: grew membership 80 to 200, owning budget and programming.

TECHNICAL SKILLS

Languages: Python, C++, C#, SQL, TypeScript, JavaScript, Java, Ruby, HTML/CSS
Frameworks & Infrastructure: .NET, ASP.NET MVC, React, Node.js, Rails, REST APIs, Docker, Kubernetes, Linux, CI/CD, Azure, AWS
Data & Quant: PostgreSQL, advanced SQL, Monte Carlo simulation, statistical arbitrage, PCA, regression, NumPy, pandas, scikit-learn